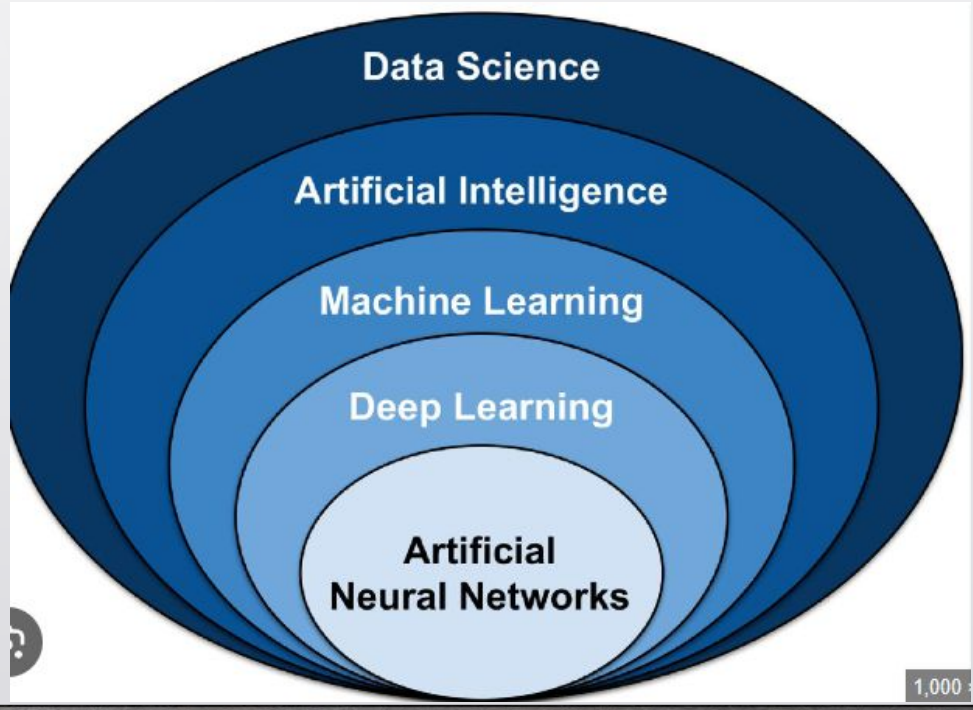


DATA SCIENCE



1. **AI** is the parent field, covering all intelligent systems.
2. **ML** is a way to make AI systems learn from data.
3. **DL** is an advanced form of ML for processing complex data.
4. **CV** focuses on understanding visual data.
5. **NLP** deals with language understanding and communication.
6. **Generative AI** is a creative offshoot of AI, creating new content like text or images.



1. Artificial Intelligence (AI):

Definition:

AI is a broad field focused on creating machines that mimic human intelligence to solve problems, learn, and make decisions.

Key Features:

- Includes systems that can reason, understand, plan, and act.
- Encompasses all areas of intelligent machine behavior.

Example:

- Self-driving cars, chatbots like Siri or Alexa, robots.

2. Machine Learning (ML):

Definition:

ML is a subset of AI that uses algorithms to allow systems to learn patterns from data and improve their performance without being explicitly programmed.

Key Features:

- Focuses on predictions and decision-making based on data.
- Learns and improves as more data becomes available.

Example:

- Netflix recommending shows, detecting spam emails.

3. Deep Learning (DL):

Definition:

Deep Learning is a specialized branch of ML that uses artificial neural networks with multiple layers to process large amounts of data and identify complex patterns.

Key Features:

- Excels at handling unstructured data like images, text, and audio.
- Requires large datasets and powerful computing resources.

Example:

- Facial recognition, voice assistants like Google Assistant.

4. Computer Vision (CV):

Definition:

Computer Vision is a field of AI focused on enabling machines to interpret and understand visual data, like images and videos.

Key Features:

- Uses deep learning to recognize objects, faces, or actions in images/videos.
- Analyzes visual inputs to extract meaningful information.

Example:

- Face unlock on smartphones, autonomous vehicle object detection.

5. Natural Language Processing (NLP):

Definition:

NLP is a branch of AI that focuses on enabling machines to understand, interpret, and generate human language (both text and speech).

Key Features:

- Handles language-related tasks like sentiment analysis, translation, and text summarization.
- Bridges communication between humans and machines.

Example:

- Chatbots like ChatGPT, Google Translate.

6. Generative AI:

Definition:

Generative AI is a type of AI that creates new content, such as text, images, audio, or videos, by learning patterns from existing data.

Key Features:

- Uses deep learning models like Generative Adversarial Networks (GANs) and Transformers.
- Focuses on creating original outputs, often resembling real-world data.

Example:

- AI-generated art, text generation (like this conversation), deepfake videos.

Feature	AI	ML	DL	CV	NLP
Scope	Broad field (umbrella term)	A part of AI	A specialized part of ML	Focuses on images/videos	Focuses on language
Goal	Mimic human intelligence	Learn from data	Learn deeper patterns	Understand visual data	Understand text/speech
Examples	Virtual assistants	Spam filters, recommendations	Self-driving cars	Face detection, object tracking	Language translation, chatbots
Data Type Processed	Varies	Tabular or structured data	Large, unstructured datasets	Images and videos	Text or spoken language
Technology Used	General AI methods	Algorithms like regression	Neural networks (deep layers)	Image processing algorithms	Text analysis algorithms

THANK YOU

